

S.C.O.U.T. Senior Design: Task Plan, Gantt Chart and WBS

Santa Clara Oceanic Utilities Transmitter · whole team · detailed plan Oct 6 – Oct 18, 2026, all known fall tasks through Jan 15, 2027 · built 2026-10-06 from docs/planning/wbs-tasks.csv (synced from Linear)

Team and ownership

- **John Ryan Myrdal (GENG)** — field and mechanical: buoy structure, deployment. 40 tasks.
- **Isabella Rodriguez (ECEN)** — hardware: electrical design, power, bench tests. 19 tasks.
- **David Chousal Cantu (CSEN)** — software: firmware, data pipeline, shore station. 10 tasks.
- Advisors: Jes Kuczenski, Hoeseok Yang (ECEN), Navid Shaghghi.

Milestones and external deadlines

- **Oct 08, 2026** — Component list delivered; unblocks housing dimensions (*Milestone*)
- **Oct 09, 2026** — Bench parts arrive (assumed Fri) (*Milestone*)
- **Oct 16, 2026** — Phase 1 exit: subsystem bring-up complete (*Milestone*)
- **Oct 18, 2026** — SCU: Undergraduate Programs Project Funding Request due; Gantt chart due (*SCU deadline*)
- **Oct 19, 2026** — Phase 2 starts: system integration (*Milestone*)
- **Nov 25, 2026** — Phase 2 exit (*Milestone*)
- **Jan 15, 2027** — Phase 3 exit: enclosure and waterproofing (*Milestone*)
- **Feb 01, 2027** — SCU: Senior Design Conference registration closes (winter) (*SCU deadline*)

How to read this

- **WBS:** 1 Project management · 2 Electrical · 3 Firmware & software · 4 Mechanical · 5 Deployment. Each task has a WBS number and, where one exists, its Linear issue.
- **Pages 2–5:** the Oct 6–18 detail, one block per work area, with dependency arrows. **Then:** the fall overview to Jan 15, and the full table with owner, dates, duration, dependencies, outside constraints and milestones for every task.
- **Dates:** solid bars end on the issue's Linear due date (a start is Linear's when the issue has started, otherwise proposed). Dashed bars are estimates; the table marks the source of every date. Parts arrival on Fri Oct 9 is an assumption to confirm.

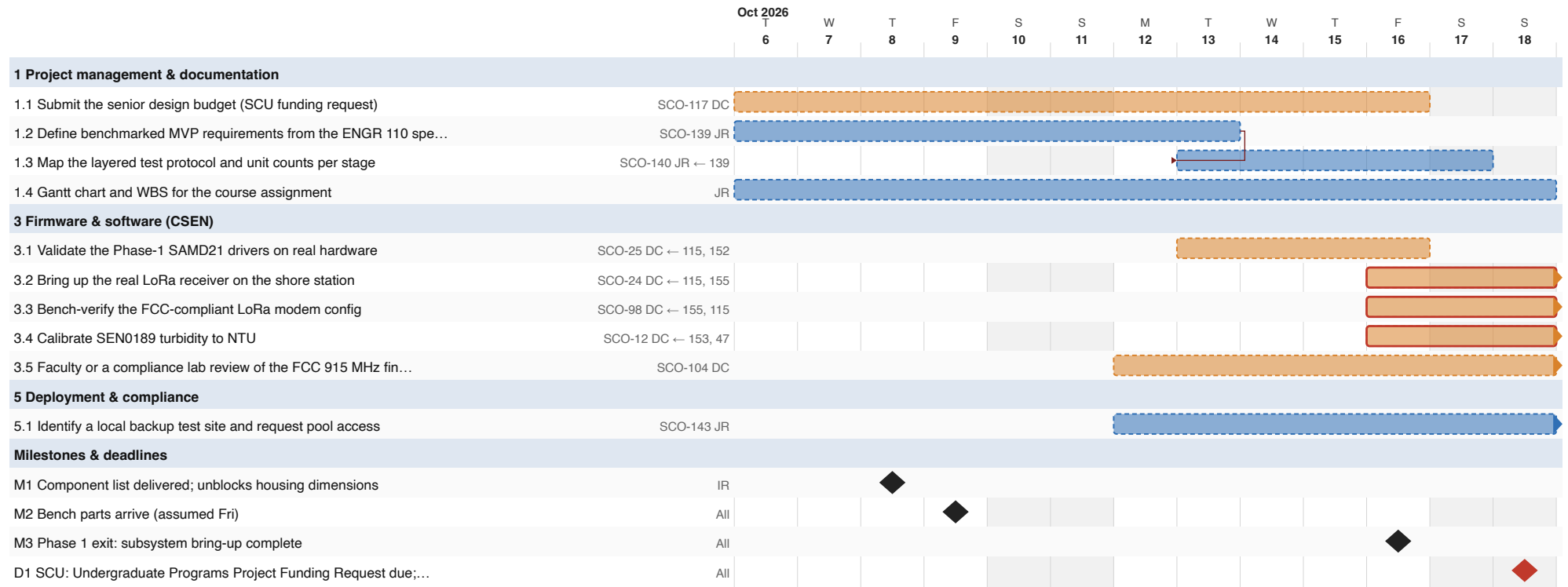
■ John Ryan Myrdal (GENG) ■ Isabella Rodriguez (ECE) ■ David Chousal Cantu (CSEN) ■ dashed = estimated dates ■ red outline = blocked / waiting
→ dependency (← lists the Linear issues a task waits for) ◆ milestone ◆ external deadline

Outside constraints that drive the plan: SCU funding request and this chart due Oct 18; weekly team meeting (Thu); parts arrival and vendor lead times; shared printer access (industrial printer, Prusa XL, lab printer); advisor and reviewer availability; saltwater soak times of 24 hours to 1 week.

Detail: Oct 6 – Oct 18, 2026 · WBS 1 / 3 / 5 / Milestones & deadlines

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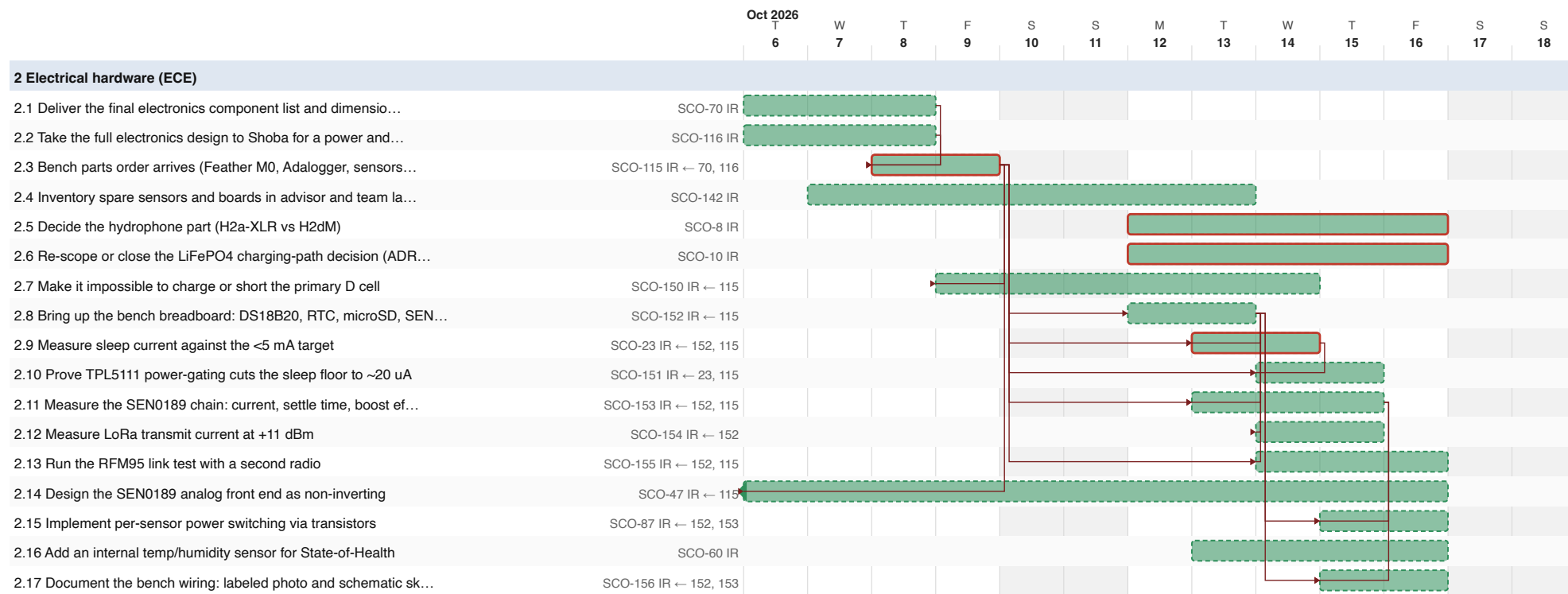
→ dependency (← lists the Linear issues a task waits for) ♦ milestone ◆ external deadline



Detail: Oct 6 – Oct 18, 2026 · WBS 2

■ John Ryan Myrdal (GENG) ■ Isabella Rodriguez (ECE) ■ David Chousal Cantu (CSEN) ■ dashed = estimated dates ■ red outline = blocked / waiting

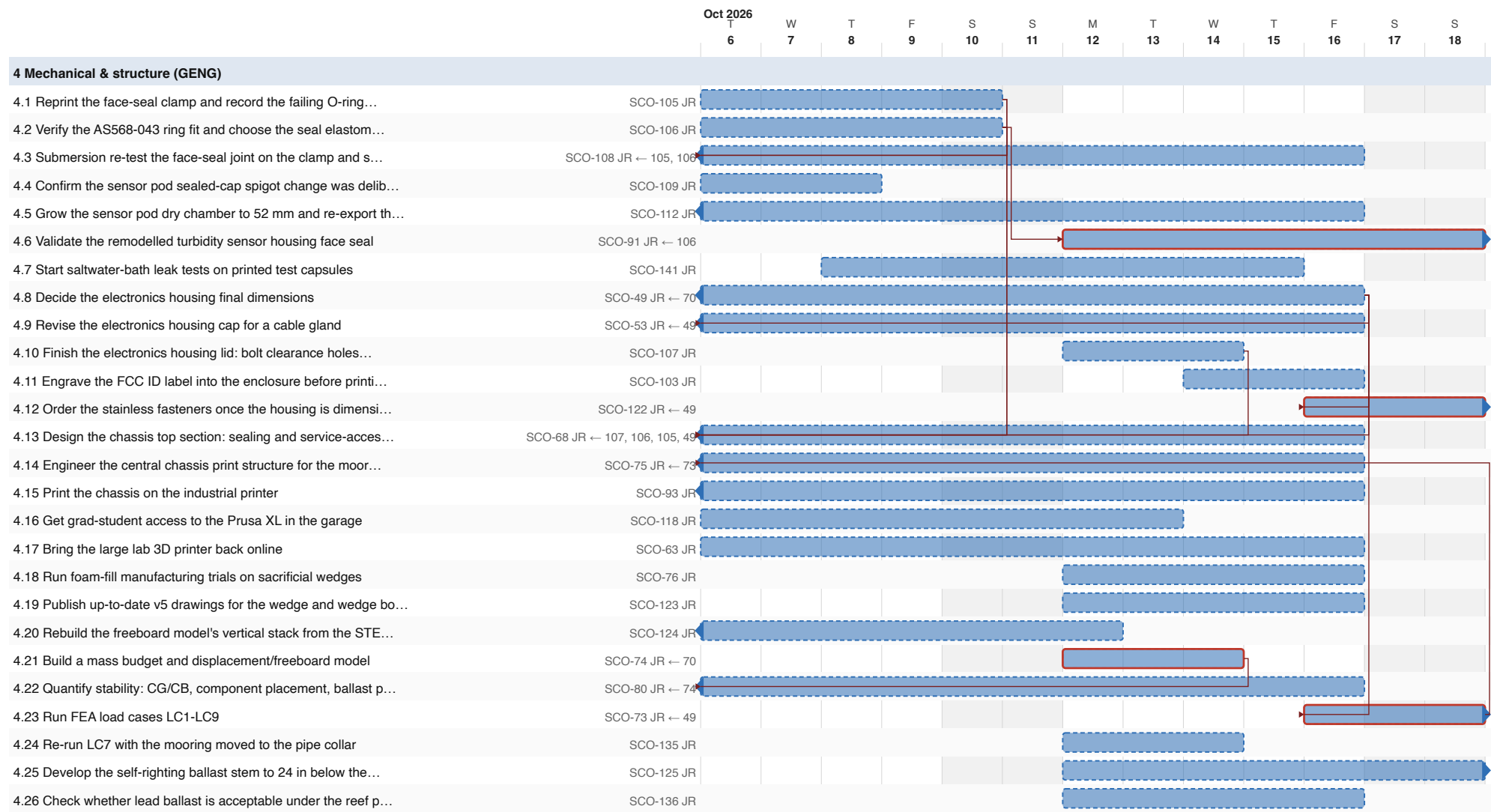
→ dependency (← lists the Linear issues a task waits for) ♦ milestone ◆ external deadline



Detail: Oct 6 – Oct 18, 2026 · WBS 4

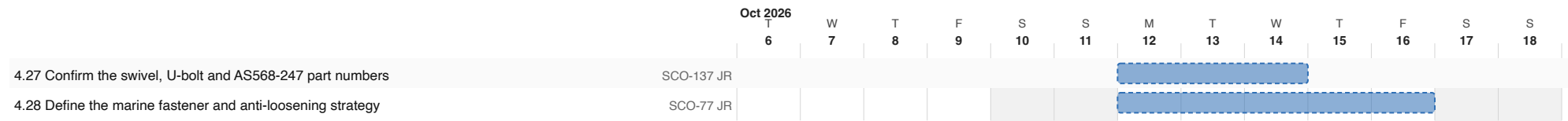
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Detail: Oct 6 – Oct 18, 2026 · WBS

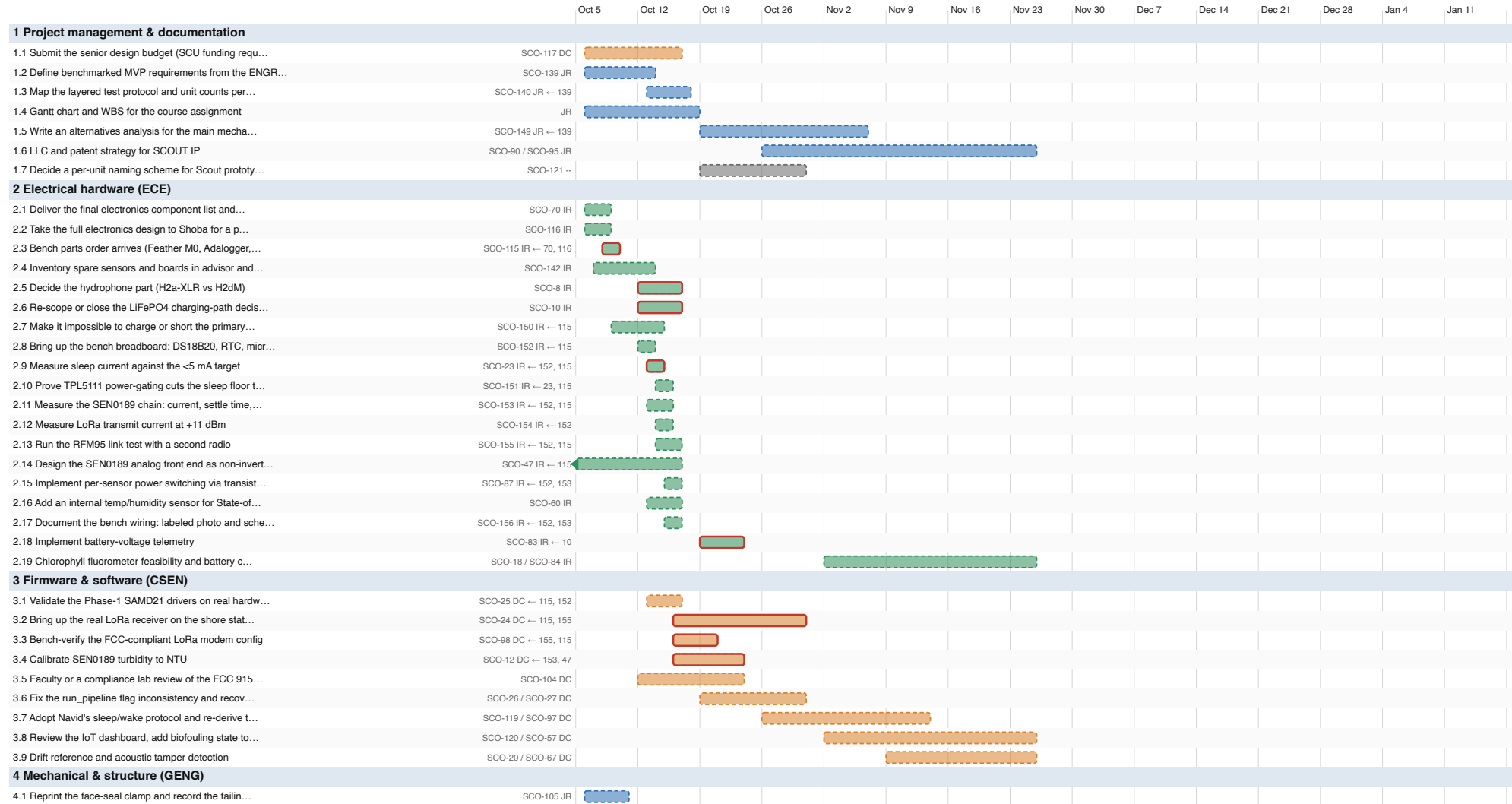
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 ◆ milestone
 ◆ external deadline



Fall overview: Oct 2026 – Jan 2027 (week columns)

 John Ryan Myrdal (GENG)
 Isabella Rodriguez (ECE)
 David Chousal Cantu (CSEN)
 dashed = estimated dates
 red outline = blocked / waiting

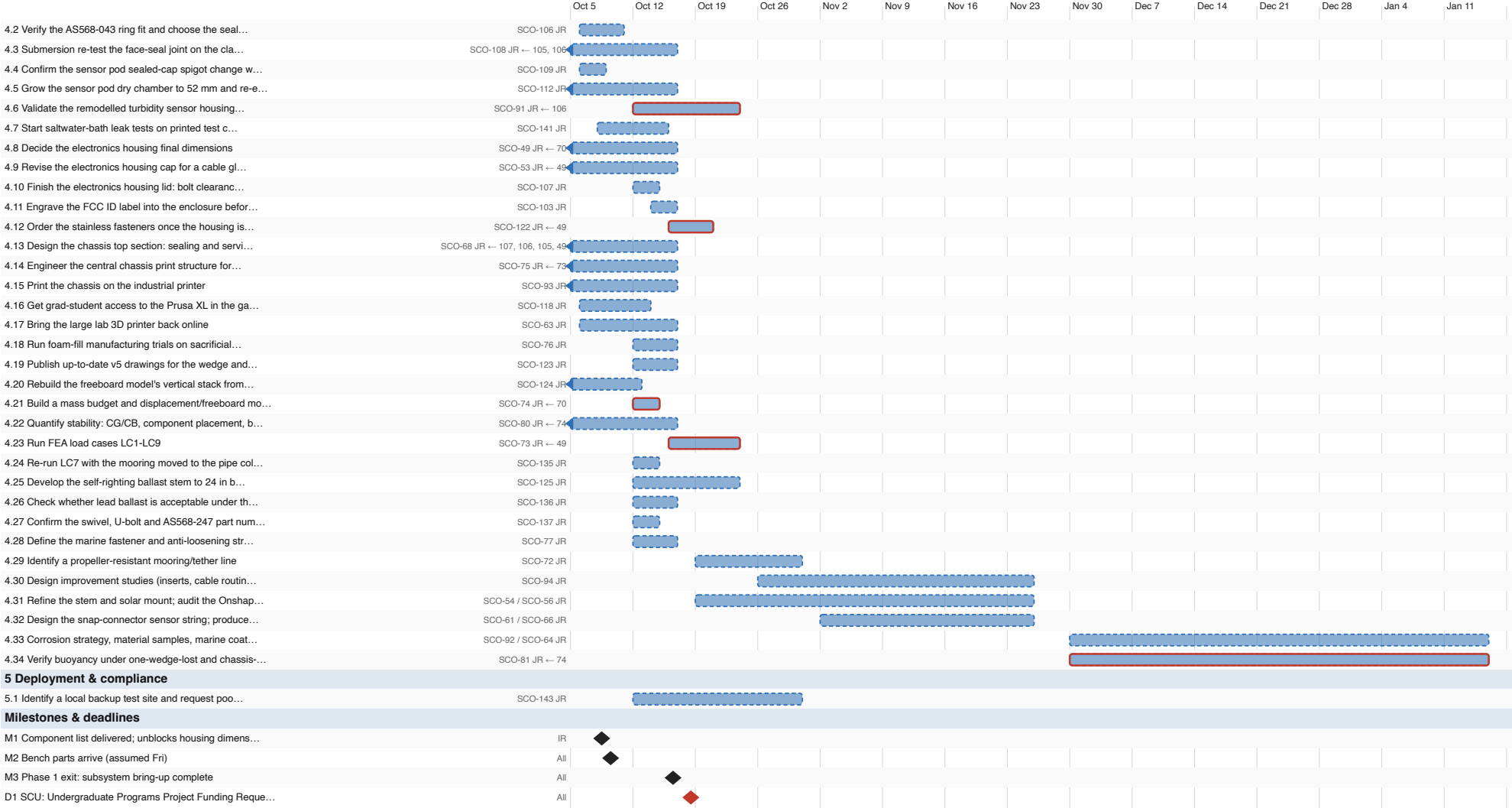
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Fall overview: Oct 2026 – Jan 2027 (week columns) (cont.)

 John Ryan Myrdal (GENG)
 Isabella Rodriguez (ECE)
 David Chousal Cantu (CSEN)
 dashed = estimated dates
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→ dependency (← lists the Linear issues a task waits for) ◆ milestone ◆ external deadline



Fall overview: Oct 2026 – Jan 2027 (week columns) (cont.)

■ John Ryan Myrdal (GENG)
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Task table (all columns)

WBS	Linear	Task	Owner	Phase	Status	Start	End	Days	Depends on	Outside constraint	Milestone / deadline	Date source
1 Project management & documentation												
1.1	SCO-117	Submit the senior design budget (SCU funding request)	David	Phase 1	Todo	2026-10-06	2026-10-16	11		Lab budget was due the week of 2026-09-28 (overdue); SCU funding request deadline Oct 18; must match the parts order	Funding request submitted	Linear due
1.2	SCO-139	Define benchmarked MVP requirements from the ENGR 110 specs	John	Phase 1	Todo	2026-10-06	2026-10-13	8		Advisor asked for measurable benchmarks (2026-09-30 meeting)		estimated
1.3	SCO-140	Map the layered test protocol and unit counts per stage	John	Phase 1	Todo	2026-10-13	2026-10-17	5	SCO-139	Advisor asked for a layered test plan (2026-09-30 meeting)	Test protocol drafted	estimated
1.4		Gantt chart and WBS for the course assignment	John	Phase 1	In Progress	2026-10-06	2026-10-18	13		Course deadline Oct 18; whole team	Gantt submitted	derived
1.5	SCO-149	Write an alternatives analysis for the main mechanical choices	John	Phase 1	Backlog	2026-10-19	2026-11-06	19	SCO-139			estimated
1.6	SCO-90 / SCO-95	LLC and patent strategy for SCOUT IP	John	Phase 1	Backlog	2026-10-26	2026-11-25	31		Possible advisor or university IP office input		estimated
1.7	SCO-121	Decide a per-unit naming scheme for Scout prototypes	Unassigned	Phase 1	Backlog	2026-10-19	2026-10-30	12		Unassigned in Linear; needs an owner and a team decision		estimated
2 Electrical hardware (ECE)												
2.1	SCO-70	Deliver the final electronics component list and dimensions	Isabella	Phase 1	Todo	2026-10-06	2026-10-08	3		Overdue since 2026-09-04; blocks housing dimensions	Component list final	Linear due
2.2	SCO-116	Take the full electronics design to Shoba for a power and solar review	Isabella	Phase 1	Todo	2026-10-06	2026-10-08	3		Shoba's availability; Dr. Yang review follows	Design review complete	Linear due
2.3	SCO-115	Bench parts order arrives (Feather M0, Adalogger, sensors, boost, TPL5111)	Isabella	Phase 1	Waiting on Parts	2026-10-08	2026-10-09	2	SCO-70, SCO-116	Vendor shipping; ordered through Dr. Krishnan 2026-10-05; arrival date assumed	Parts arrive	Linear due
2.4	SCO-142	Inventory spare sensors and boards in advisor and team labs	Isabella	Phase 1	Todo	2026-10-07	2026-10-13	7		Needs advisor and lab access		Linear due
2.5	SCO-8	Decide the hydrophone part (H2a-XLR vs H2dM)	Isabella	Phase 1	Needs Decision	2026-10-12	2026-10-16	5		Hydrophone not in the bench order	Decision recorded	Linear due
2.6	SCO-10	Re-scope or close the LiFePO4 charging-path decision (ADR-0002)	Isabella	Phase 1	Needs Decision	2026-10-12	2026-10-16	5		Overtaken if ADR-0007 (Li-SOCl2 primary cell) is accepted; unblocks SCO-83	Decision recorded	Linear due
2.7	SCO-150	Make it impossible to charge or short the primary D cell	Isabella	Phase 1	Todo	2026-10-09	2026-10-14	6	SCO-115	Fire-safety layers must exist before any real cell; bench uses a current-limited supply		Linear due
2.8	SCO-152	Bring up the bench breadboard: DS18B20, RTC, microSD, SEN0189 wiring	Isabella	Phase 1	Todo	2026-10-12	2026-10-13	2	SCO-115	Needs the parts to arrive; bench supply and uA meter reserved	Bench bring-up complete	Linear due
2.9	SCO-23	Measure sleep current against the <5 mA target	Isabella	Phase 1	Waiting on Parts	2026-10-13	2026-10-14	2	SCO-152, SCO-115	uA-range multimeter; 3.6 V current-limited supply		Linear due
2.10	SCO-151	Prove TPL5111 power-gating cuts the sleep floor to ~20 uA	Isabella	Phase 1	Todo	2026-10-14	2026-10-15	2	SCO-23, SCO-115	Firmware cold-boot support from David (SCO-25)		Linear due
2.11	SCO-153	Measure the SEN0189 chain: current, settle time, boost efficiency, inrush, polarity	Isabella	Phase 1	Todo	2026-10-13	2026-10-15	3	SCO-152, SCO-115	Oscilloscope for inrush		Linear due
2.12	SCO-154	Measure LoRa transmit current at +11 dBm	Isabella	Phase 1	Todo	2026-10-14	2026-10-15	2	SCO-152	1 ohm shunt; re-check after SCO-98		Linear due
2.13	SCO-155	Run the RFM95 link test with a second radio	Isabella	Phase 1	Todo	2026-10-14	2026-10-16	3	SCO-152, SCO-115	Second radio is not in the 10-05 order; lab stock or a new order	Radio link proven	Linear due
2.14	SCO-47	Design the SEN0189 analog front end as non-inverting	Isabella	Phase 1	In Progress	2026-09-01	2026-10-16	46	SCO-115	Polarity must not invert; bench check after parts arrive; Design started 2026-09-01; bench verification follows SCO-115 and SCO-153		Linear due
2.15	SCO-87	Implement per-sensor power switching via transistors	Isabella	Phase 1	Todo	2026-10-15	2026-10-16	2	SCO-152, SCO-153	V1-required on a primary cell		Linear due

2.16	SCO-60	Add an internal temp/humidity sensor for State-of-Health	Isabella	Phase 1	Todo	2026-10-13	2026-10-16	4				Linear due
2.17	SCO-156	Document the bench wiring: labeled photo and schematic sketch to /hardware	Isabella	Phase 1	Todo	2026-10-15	2026-10-16	2	SCO-152, SCO-153	Phase 1 exit item	Phase 1 hardware exit	Linear due
2.18	SCO-83	Implement battery-voltage telemetry	Isabella	Phase 1	Blocked	2026-10-19	2026-10-23	5	SCO-10	Depends on the battery chemistry decision		Linear due
2.19	SCO-18 / SCO-84	Chlorophyll fluorometer feasibility and battery connector polarity check	Isabella	Phase 2	Deferred	2026-11-02	2026-11-25	24		Deferred in Linear; not this version		estimated
3 Firmware & software (CSEN)												
3.1	SCO-25	Validate the Phase-1 SAMD21 drivers on real hardware	David	Phase 1	Todo	2026-10-13	2026-10-16	4	SCO-115, SCO-152	Needs the Feather M0 from the bench order	Drivers validated on hardware	Linear due
3.2	SCO-24	Bring up the real LoRa receiver on the shore station	David	Phase 1	Waiting on Parts	2026-10-16	2026-10-30	15	SCO-115, SCO-155	Linear due 10-16 moved to 10-30: shore parts (Feather S3, LoRa module) are not ordered yet		Linear due
3.3	SCO-98	Bench-verify the FCC-compliant LoRa modem config	David	Phase 1	Blocked	2026-10-16	2026-10-20	5	SCO-155, SCO-115	Config is merged; only bench verification remains		estimated
3.4	SCO-12	Calibrate SEN0189 turbidity to NTU	David	Phase 1	Blocked	2026-10-16	2026-10-23	8	SCO-153, SCO-47	Blocked since 2026-08-17; needs the SEN0189 front end (SCO-47) and chain measurements (SCO-153); formazin standards; due moved 10-16 to 10-23		Linear due
3.5	SCO-104	Faculty or a compliance lab review of the FCC 915 MHz findings	David	Phase 1	Todo	2026-10-12	2026-10-23	12		Reviewer availability		estimated
3.6	SCO-26 / SCO-27	Fix the run_pipeline flag inconsistency and recover the missing Sesoko recording	David	Phase 1	Backlog	2026-10-19	2026-10-30	12		Recording may need the original data source		estimated
3.7	SCO-119 / SCO-97	Adopt Navid's sleep/wake protocol and re-derive the packet ceiling	David	Phase 2	Backlog	2026-10-26	2026-11-13	19				estimated
3.8	SCO-120 / SCO-57 / SCO-58	Review the IoT dashboard, add biofouling state to the Fleet page, improve the PNG dashboard	David	Phase 2	Backlog	2026-11-02	2026-11-25	24			Phase 2 software exit	estimated
3.9	SCO-20 / SCO-67	Drift reference and acoustic tamper detection	David	Phase 2	Deferred	2026-11-09	2026-11-25	17		Deferred in Linear; not this version		estimated
4 Mechanical & structure (GENG)												
4.1	SCO-105	Reprint the face-seal clamp and record the failing O-ring tolerance	John	Phase 1	Todo	2026-10-06	2026-10-10	5		Printer access		estimated
4.2	SCO-106	Verify the AS568-043 ring fit and choose the seal elastomer before ordering	John	Phase 1	Todo	2026-10-06	2026-10-10	5		Decision: silicone elastomer (2026-09-08)		estimated
4.3	SCO-108	Submersion re-test the face-seal joint on the clamp and sensor pod	John	Phase 1	In Progress	2026-10-01	2026-10-16	16	SCO-105, SCO-106	24 hr soaks; RR-14 result 10-07; RR-15 (6 bolts) after; Coupon re-tests (RR-14, RR-15) run now; the clamp re-test follows SCO-105 and SCO-106	Face seal passes a 24 hr soak	estimated
4.4	SCO-109	Confirm the sensor pod sealed-cap spigot change was deliberate	John	Phase 1	Todo	2026-10-06	2026-10-08	3				estimated
4.5	SCO-112	Grow the sensor pod dry chamber to 52 mm and re-export the penetration and flood chamber	John	Phase 1	In Progress	2026-09-14	2026-10-16	33		Epoxy-potted lead penetration		estimated
4.6	SCO-91	Validate the remodelled turbidity sensor housing face seal	John	Phase 1	Blocked	2026-10-12	2026-10-23	12	SCO-106			estimated
4.7	SCO-141	Start saltwater-bath leak tests on printed test capsules	John	Phase 1	Todo	2026-10-08	2026-10-15	8		1 week soak at about 35 ppt		estimated
4.8	SCO-49	Decide the electronics housing final dimensions	John	Phase 1	In Progress	2026-08-18	2026-10-16	60	SCO-70	Needs Isabella's component list; drawing DFM Assembly 2 drafted 2026-10-06; Started 2026-08-18; the final decision waits on SCO-70	Housing dimensions final	Linear due
4.9	SCO-53	Revise the electronics housing cap for a cable gland	John	Phase 1	In Progress	2026-08-16	2026-10-16	62	SCO-49	Started 2026-08-16; gland placement follows SCO-49 geometry; gland sizing and tolerance tests outstanding		Linear due
4.10	SCO-107	Finish the electronics housing lid: bolt clearance holes and marking wording	John	Phase 1	Todo	2026-10-12	2026-10-14	3				estimated

4.11	SCO-103	Engrave the FCC ID label into the enclosure before printing	John	Phase 1	Todo	2026-10-14	2026-10-16	3		Engraved before the final print (related to SCO-107 and SCO-93, not blocked by them)		estimated
4.12	SCO-122	Order the stainless fasteners once the housing is dimensioned	John	Phase 1	Blocked	2026-10-16	2026-10-20	5	SCO-49	Vendor lead time; ordered only after SCO-49 fixes the housing dimensions		estimated
4.13	SCO-68	Design the chassis top section: sealing and service-access port	John	Phase 1	In Progress	2026-09-03	2026-10-16	44	SCO-107, SCO-106, SCO-105, SCO-49			Linear due
4.14	SCO-75	Engineer the central chassis print structure for the mooring load path	John	Phase 1	In Progress	2026-08-21	2026-10-16	57	SCO-73	Linear lists SCO-73 as blocking SCO-75, yet SCO-75 is In Progress and ends before SCO-73 does; John to confirm the relation or the dates		Linear due
4.15	SCO-93	Print the chassis on the industrial printer	John	Phase 1	In Progress	2026-09-30	2026-10-16	17		Industrial printer window at work; about 35 hr print of the v5 design (later changes feed a reprint)	Chassis printed	estimated
4.16	SCO-118	Get grad-student access to the Prusa XL in the garage	John	Phase 1	Todo	2026-10-06	2026-10-13	8		Needs a grad student's permission		estimated
4.17	SCO-63	Bring the large lab 3D printer back online	John	Phase 1	Backlog	2026-10-06	2026-10-16	11		Lab technician or advisor help		Linear due
4.18	SCO-76	Run foam-fill manufacturing trials on sacrificial wedges	John	Phase 1	Todo	2026-10-12	2026-10-16	5		Printer access for sacrificial wedges		Linear due
4.19	SCO-123	Publish up-to-date v5 drawings for the wedge and wedge bottom	John	Phase 1	Todo	2026-10-12	2026-10-16	5				estimated
4.20	SCO-124	Rebuild the freeboard model's vertical stack from the STEP geometry	John	Phase 1	In Progress	2026-09-29	2026-10-12	14		v7 STEP committed 2026-10-06		estimated
4.21	SCO-74	Build a mass budget and displacement/freeboard model	John	Phase 1	Blocked	2026-10-12	2026-10-14	3	SCO-70	Linear due moved 10-16 to 10-14 so the stability work (SCO-80) can follow; needs component masses from SCO-70	Freeboard model complete	Linear due
4.22	SCO-80	Quantify stability: CG/CB, component placement, ballast provision	John	Phase 1	In Progress	2026-08-21	2026-10-16	57	SCO-74	Started 2026-08-21; final numbers follow the SCO-74 mass budget		Linear due
4.23	SCO-73	Run FEA load cases LC1-LC9	John	Phase 1	Blocked	2026-10-16	2026-10-23	8	SCO-49	Blocked by SCO-49; Linear due unset		estimated
4.24	SCO-135	Re-run LC7 with the mooring moved to the pipe collar	John	Phase 1	Todo	2026-10-12	2026-10-14	3				estimated
4.25	SCO-125	Develop the self-righting ballast stem to 24 in below the keel	John	Phase 1	Todo	2026-10-12	2026-10-23	12		Lead ballast permit question (SCO-136)		estimated
4.26	SCO-136	Check whether lead ballast is acceptable under the reef permit	John	Phase 1	Todo	2026-10-12	2026-10-16	5		Permit authority answer		estimated
4.27	SCO-137	Confirm the swivel, U-bolt and AS568-247 part numbers	John	Phase 1	Todo	2026-10-12	2026-10-14	3				estimated
4.28	SCO-77	Define the marine fastener and anti-loosening strategy	John	Phase 1	Backlog	2026-10-12	2026-10-16	5				Linear due
4.29	SCO-72	Identify a propeller-resistant mooring/tether line	John	Phase 1	Backlog	2026-10-19	2026-10-30	12				estimated
4.30	SCO-94 / 144 / 145 / 146 / 147 / 148	Design improvement studies (inserts, cable routing, modular rod, split chassis, float-free, manufacturability)	John	Phase 1	Backlog	2026-10-26	2026-11-25	31				estimated
4.31	SCO-54 / SCO-56	Refine the stem and solar mount; audit the Onshape Initial Frame folder	John	Phase 1	Backlog	2026-10-19	2026-11-25	38			Phase 2 mechanical exit	Linear due
4.32	SCO-61 / SCO-66	Design the snap-connector sensor string; produce an exploded-view animation	John	Phase 2	Backlog	2026-11-02	2026-11-25	24				Linear due
4.33	SCO-92 / SCO-64 / SCO-78 / SCO-79	Corrosion strategy, material samples, marine coating, biofouling features	John	Phase 3	Todo / Backlog	2026-11-30	2027-01-15	47		Winter break; printer access	Phase 3 exit	estimated
4.34	SCO-81	Verify buoyancy under one-wedge-lost and chassis-flooded cases	John	Phase 3	Blocked	2026-11-30	2027-01-15	47	SCO-74			Linear due
5 Deployment & compliance												
5.1	SCO-143	Identify a local backup test site and request pool access	John	Phase 1	Backlog	2026-10-12	2026-10-30	19		Site owner's permission		estimated
Milestones & deadlines												

M1		Component list delivered; unblocks housing dimensions	Isabella	Phase 1		2026-10-08	2026-10-08	1			Component list final	estimated
M2		Bench parts arrive (assumed Fri)	All	Phase 1		2026-10-09	2026-10-09	1		Assumption; confirm with Isabella	Parts arrive	Linear
M3		Phase 1 exit: subsystem bring-up complete	All	Phase 1		2026-10-16	2026-10-16	1		Phase 1 project window	Phase 1 exit	estimated
D1		SCU: Undergraduate Programs Project Funding Request due; Gantt chart due	All	Phase 1		2026-10-18	2026-10-18	1		SCU Senior Design Resources page	Funding request and Gantt submitted	Linear
M4		Phase 2 starts: system integration	All	Phase 2		2026-10-19	2026-10-19	1				estimated
M5		Phase 2 exit	All	Phase 2		2026-11-25	2026-11-25	1			Phase 2 exit	estimated
M6		Phase 3 exit: enclosure and waterproofing	All	Phase 3		2027-01-15	2027-01-15	1			Phase 3 exit	estimated
D2		SCU: Senior Design Conference registration closes (winter)	All	Phase 4		2027-02-01	2027-02-01	1		SCU Senior Design Resources page		Linear

Date source: **Linear due** = the end date equals the issue's Linear due date; **estimated** = proposed from the issue text, parts timing or the phase window and awaiting the owner's confirmation; **derived** = set by this assignment. Days are calendar days, inclusive.